

THOMAS CHENG

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TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, HTML/CSS, C, C++, Rust, Haskell, Java, Bash, SQL
Frameworks/Libraries: Pydantic, Sklearn, FastText, Node.js, Express, React, JUnit, Jest, Bootstrap, Zod
Developer Tools: Git, Docker, AWS EC2

TECHNICAL PROJECTS

MCP-Enabled Task System (MTS) | MCP, Node.js, Copilot, MongoDB, TypeScript, Python

- Engineered a cross-stack architecture with a TypeScript MCP service for AI Interaction and a Python display interface for live task rendering.
- Developed an MCP server using Node.js and MongoDB to allow the use of natural language to perform CRUD operations through VSCode Copilot.
- Designed a Python interface rendering tasks onto an LED dashboard via the local network.

Data Synthesis & Automation Engine (song-values) | Python, Pydantic, Pandas, Numpy, Scikit-learn, FastText

- Engineered an automated data pipeline to synthesize high-fidelity, niche datasets for model training, overcoming data scarcity limitations.
- Enforced system reliability by implementing Pydantic to validate AI-generated outputs, ensuring strict schema compliance and data integrity for downstream pipelines.
- Managed the full model lifecycle, including data preprocessing, automated training, and validation, ensuring the final model met performance benchmarks for production-level accuracy.

PCBee | Node.js, Express, Bootstrap, Jest, OAuth

- Automated the software lifecycle by integrating a CI/CD pipeline with GitHub Actions and Jest to maintain high-quality code standards and reliable deployments.
- Streamlined user requirement gathering by integrating an AI chatbot, facilitating complex data retrieval and analysis through intelligent system interaction.
- Optimized development workflows using Agile methodologies to deliver consistent updates and ensure continuous reliability for end-user features.

WORK EXPERIENCE

Computer Science Tutor

Sept 2023 – Present

Self-Employed

Vancouver, BC

- Maintained Students' grades above 80%, with a mastery of data structures, algorithms, and software engineering.
- Designed comprehensive study plans based on documenting students' growth.
- Coached students in unit testing and debugging practices, which allow students to develop sustainable coding habits.

Sales Representative

Sept 2022 – Jan 2026

Under Armour CA

Richmond, BC

- Led foundational training for new hires, showcasing ability to convey complex concepts clearly and understandably.
- Thrived in fast-paced environment during holiday seasons, translating the ability to handle sustained large customer volume into an efficient work ethic.

EDUCATION

Simon Fraser University

Burnaby, BC

Bachelor of Science in Computer Science

Sept 2024 – Apr 2027

Relevant Coursework: Data Structures and algorithms, Intro to Artificial Intelligence, Computer Systems, Database Systems

Langara College

Vancouver, BC

Associate of Science in Computer Science

Jan 2022 – May 2024

Relevant Coursework: Software Practice, Full Stack Web Dev, Data Structures, Unix tools, Object Oriented Programming